

Synthetic Data from Industrial Sensor Monitoring

Overview

This collection contains five datasets that simulate sensor readings from different industrial production lines. The data represents measurements of temperature, pressure, and in some cases, elapsed time of industrial machines, with records of normal operating conditions and potential anomalies.

Data Structure

Common Format

All datasets are in CSV format with the following common fields:

- **timestamp**: Date and time of measurement (format YYYY-MM-DD HH:MM:SS)
- **temperature/Temperature**: Temperature value in arbitrary units
- **pressure/Pressure**: Pressure value in arbitrary units
- **label**: Binary indicator (0 = normal operation, 1 = anomaly)

Some datasets include an additional field:

- **elapsed_time/Elapsed_time**: Machine runtime in arbitrary units

Dataset Descriptions

1. LineA_Stable_10K.csv

- **Period**: May 2025
- **Fields**: timestamp, Temperature, pressure, elapsed_time, label
- **Characteristics**: Low variability in temperature and pressure, larger dataset with 10,000 records
- **Size**: 757.50KB
- **Notes**: Most stable production line with consistent readings

2. LineB_Flux.csv

- **Period:** April 2025
- **Fields:** timestamp, temperature, pressure, Elapsed_time, label
- **Characteristics:** Medium variability in temperature and pressure
- **Size:** 381.55KB
- **Notes:** Production line with moderate fluctuations

3. LineC_Turbulent.csv

- **Period:** March 2025
- **Fields:** timestamp, Temperature, pressure, label
- **Characteristics:** High variability in temperature, medium variability in pressure
- **Size:** 288.11KB
- **Notes:** Production line with turbulent conditions and significant fluctuations

4. LineD_SpikeControl.csv

- **Period:** February 2025
- **Fields:** timestamp, temperature, Pressure, label
- **Characteristics:** High variability in temperature, low variability in pressure
- **Size:** 288.18KB
- **Notes:** Production line with controlled pressure but temperature spikes

5. LineE_SmoothRun.csv

- **Period:** January 2025
- **Fields:** timestamp, Temperature, pressure, label
- **Characteristics:** Low variability in both temperature and pressure
- **Size:** 288.17KB
- **Notes:** Production line with smooth operation and minimal fluctuations

Data Statistics

Dataset	Temperature Range	Pressure Range	Elapsed Time Range	% of Anomalies
LineA_Stable_10K	~179-180	~159-160	~34-35	< 1%
LineB_Flux	~188-191	~118-122	~19-20	< 1%
LineC_Turbulent	~196-210	~97-103	N/A	< 5%
LineD_SpikeControl	~196-202	~97-102	N/A	< 5%
LineE_SmoothRun	~199-200	~99-100	N/A	0%

Data Generation Methodology

The data were synthetically generated to simulate real operational conditions of industrial production lines. Anomalies were introduced to represent potential failures or abnormal operating conditions. Each dataset represents a different production line with specific characteristics regarding stability and variability.

Suggested Applications

- Anomaly detection in industrial environments
- Machine failure prediction
- Time series analysis of sensor data
- Development of predictive maintenance systems
- Benchmarking machine learning algorithms for industrial IoT
- Comparative analysis of production line stability

Suggested Citation

If you use these datasets in your research, please cite as:

- Carneiro, D., Torres, D., & Peixoto, E. (2025). "Synthetic Data from Industrial Sensor Monitoring". ZENODO.

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Contact

- Davide Carneiro
- davide.r.carneiro@inesctec.pt
- Escola Superior de Tecnologia e Gestão, Instituto Politécnico do Porto, 4610-156 Felgueiras, Portugal
- INESC TEC, R. Dr. Roberto Frias, 4200-465 Porto, Portugal